Left Column: Architecture

The server is a single binary. The HTTP layer owns the routes, the handlers own the semantics, and the storage layer owns the rows.

Each layer is replaceable behind its seam. The router merges sub-routers; the handlers call services; the services touch storage through one pool. Nothing reaches across two layers at once.

The design keeps the memory envelope small. The hot path allocates bounded buffers and the pools cap their connections.

Right Column: Operations

Ops tools are verbs, not daemons. The operator runs a command, reads the exit code, and closes the terminal.

Backups are file copies. Restore is the copy run backward. The SQL migration rehearsal replays a snapshot into a scratch database before the live schema moves, so the upgrade path is measured, not hoped for.

Every alert names its remedy. A page without a runbook is a page that will page again.

Left Column: Verification

The suite is the largest artifact. Its shape is measured, its classes are derived by a rule, and the census is re-runnable.

A pin without a control arm is a check that cannot fail. The negative side ships with every positive claim.

Right Column: Economics

Latency is a budget. The recall budget is single-digit milliseconds on the edge profile, and every added stage must state its cost before it lands.

Storage is the scarcest resource on the small boxes. The vector store quantizes to int8 plus a binary sidecar and keeps the float32 rows only for the backfill window.